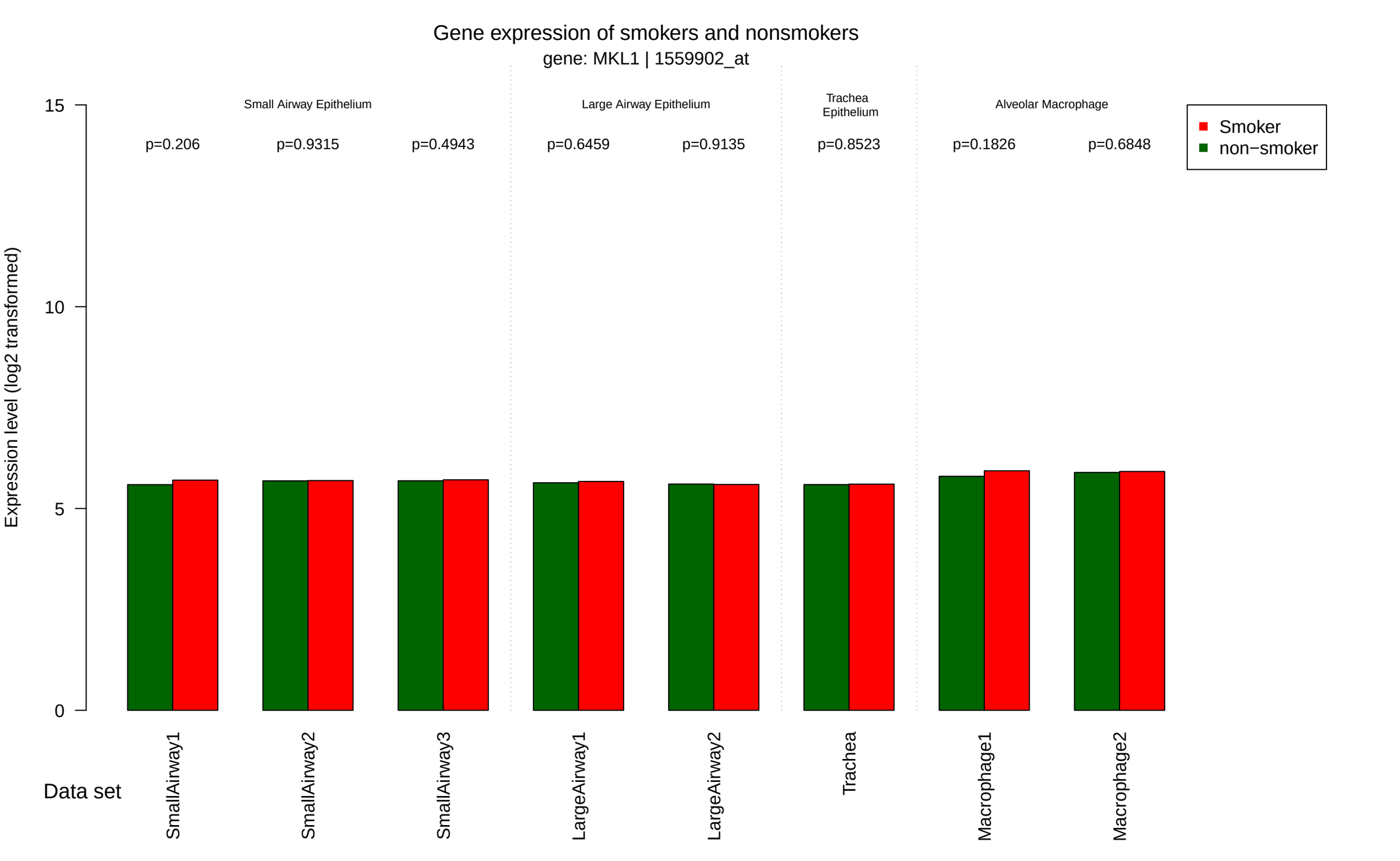
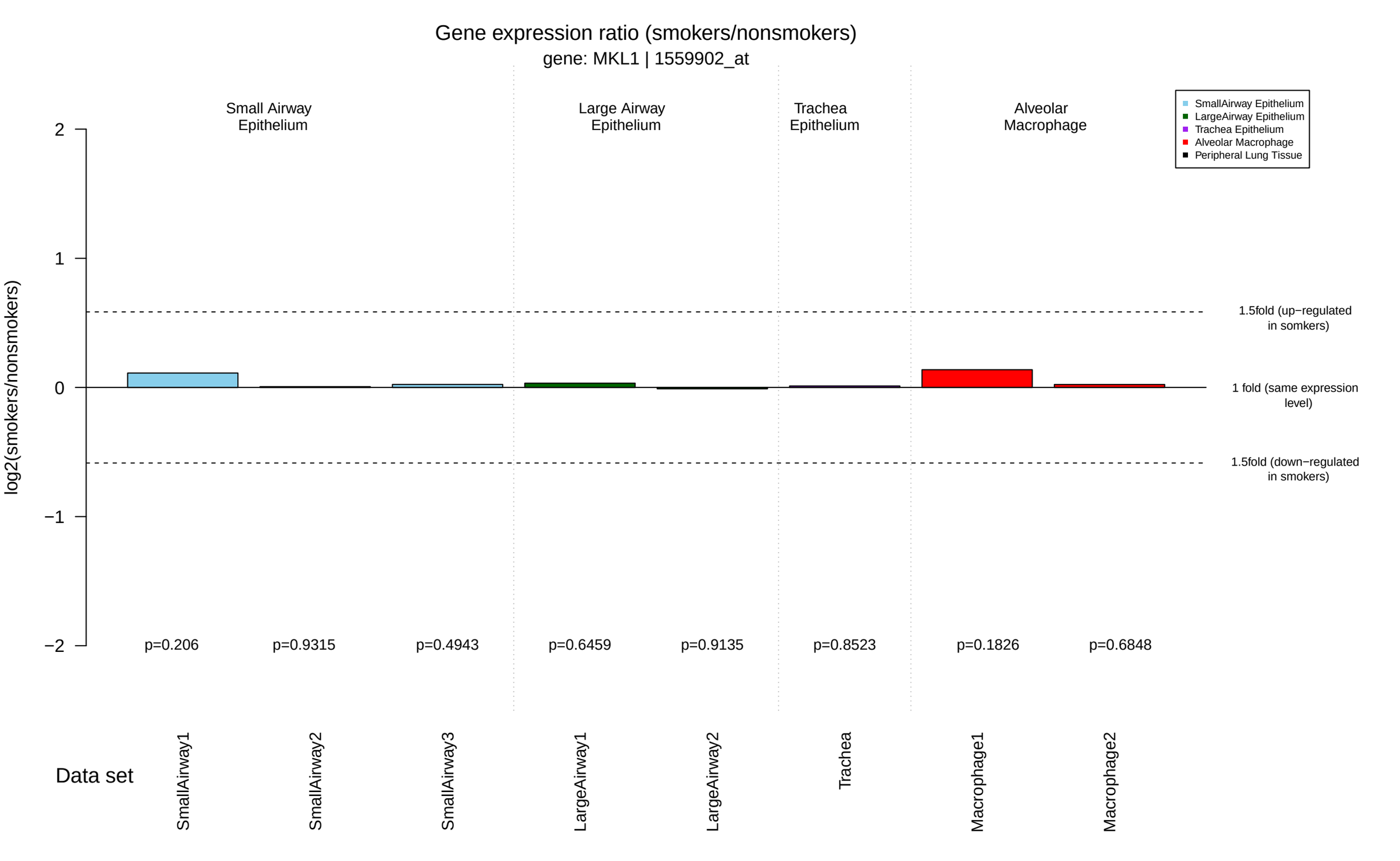
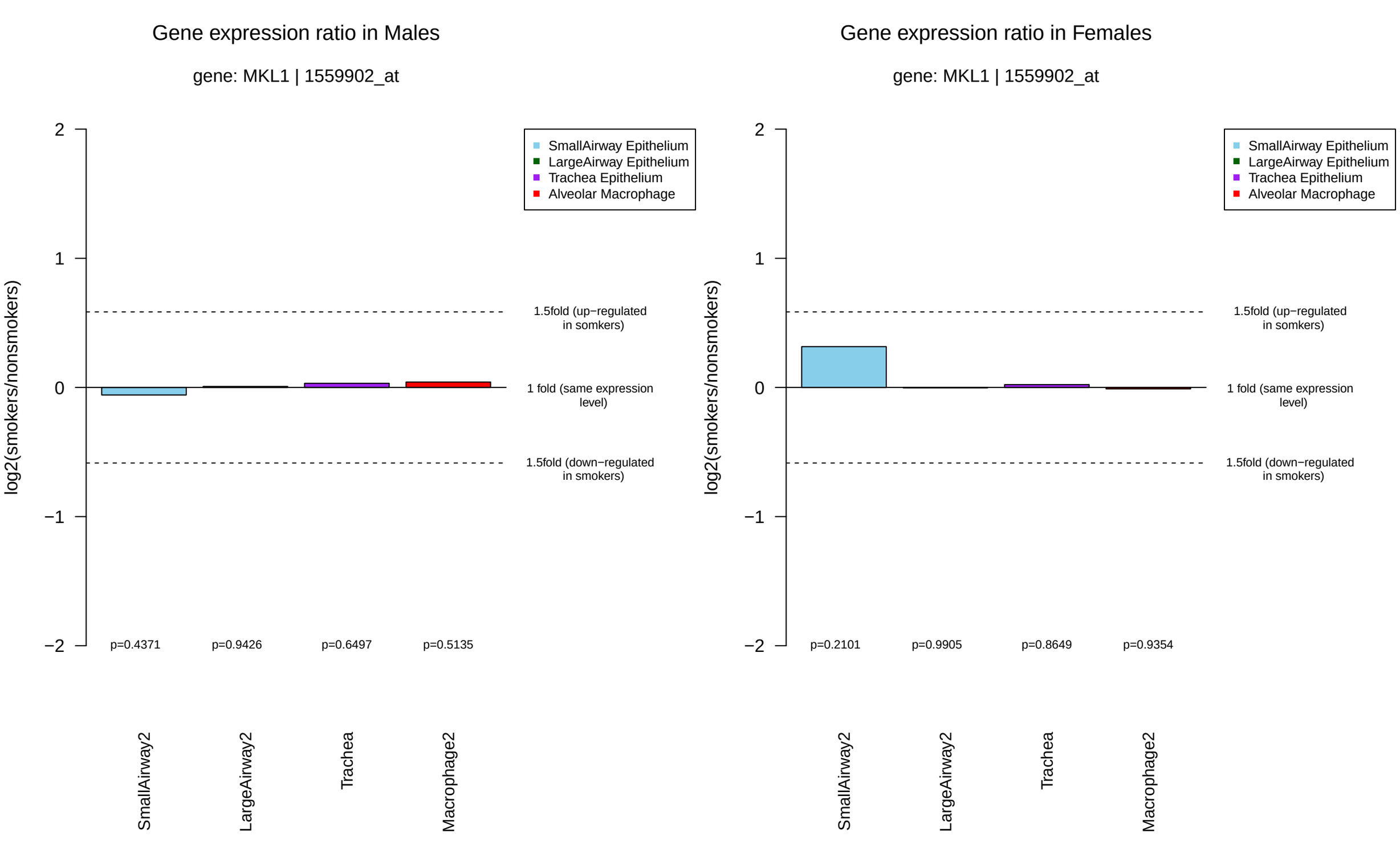
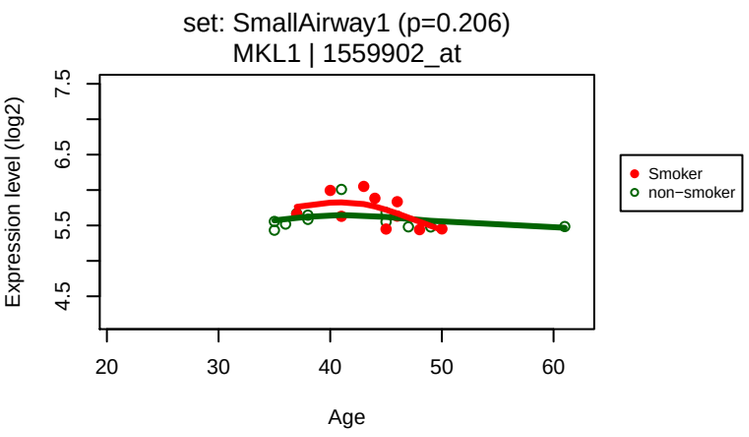
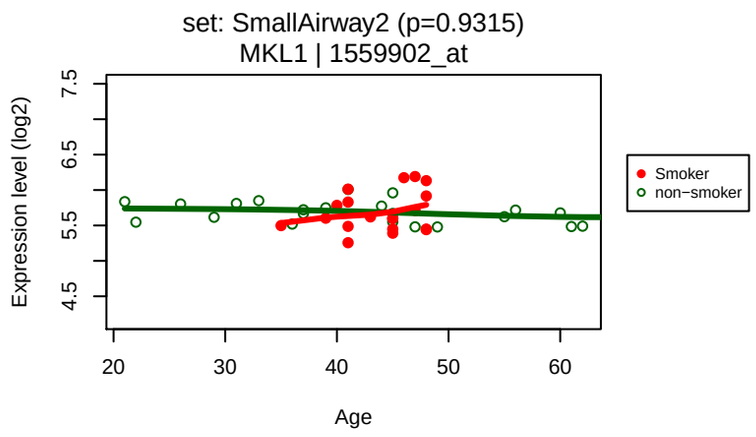
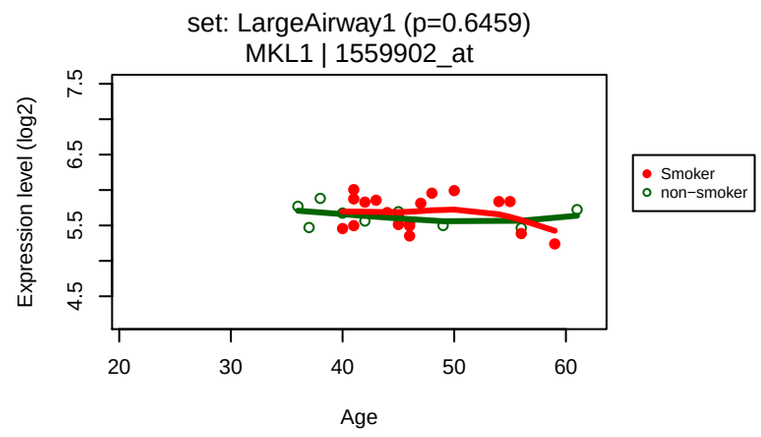

Gene expression of smokers and nonsmokers
gene: MKL1 | 1559902_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.206
p=0.9315
p=0.4943
p=0.6459
p=0.9135
p=0.8523
p=0.1826
p=0.6848
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

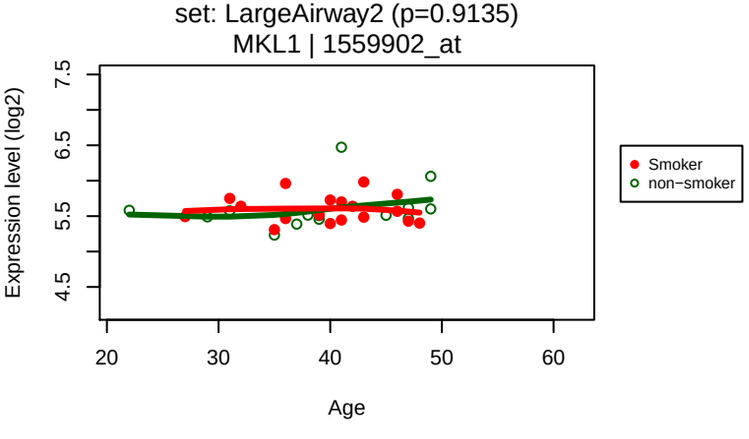

Gene expression ratio (smokers/nonsmokers)
gene: MKL1 | 1559902_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.206
p=0.9315
p=0.4943
p=0.6459
p=0.9135
p=0.8523
p=0.1826
p=0.6848
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

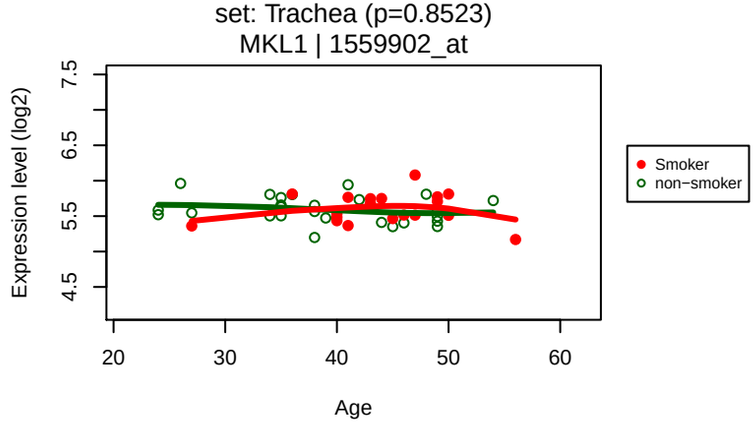

Gene expression ratio in Males
gene: MKL1 | 1559902_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4371
p=0.9426
p=0.6497
p=0.5135
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: MKL1 | 1559902_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2101
p=0.9905
p=0.8649
p=0.9354
SmallAirway2
LargeAirway2
Trachea
Macrophage2

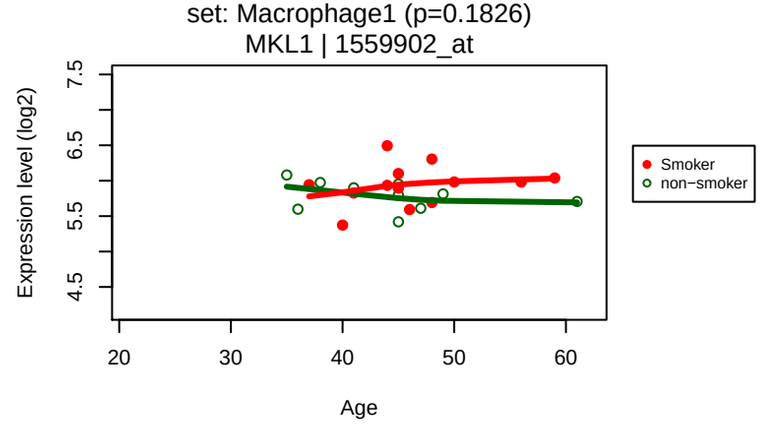

set: SmallAirway1 (p=0.206)
MKL1 | 1559902_at
Expression level (log2)
4.5
5.5
6.5
7.5
20
30
40
50
60
Age
Smoker
non−smoker

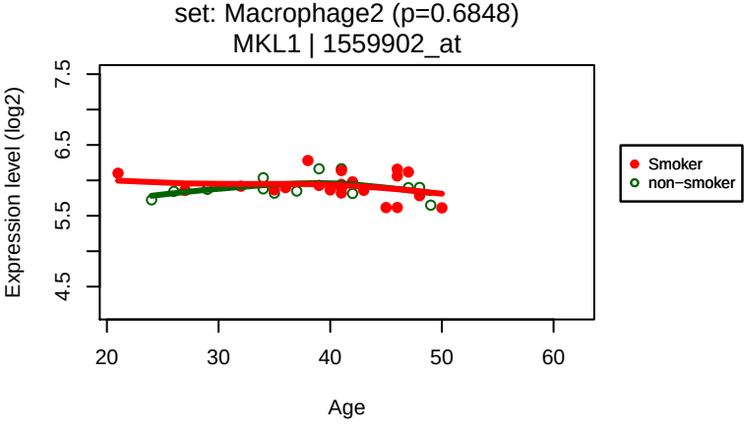

set: SmallAirway2 (p=0.9315)
MKL1 | 1559902_at
Expression level (log2)
4.5
5.5
6.5
7.5
20
30
40
50
60
Age
Smoker
non−smoker

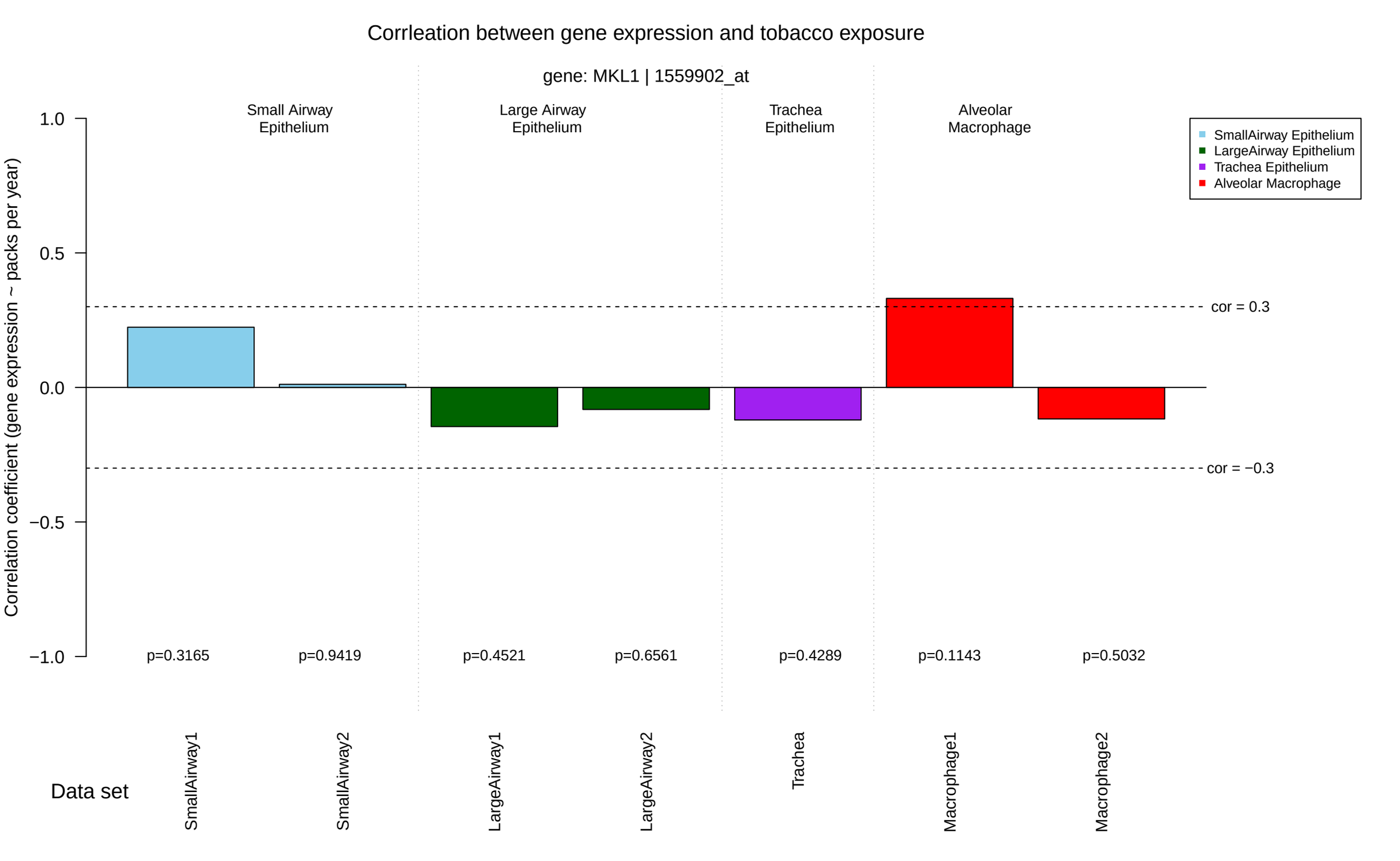

set: LargeAirway1 (p=0.6459)
MKL1 | 1559902_at
Expression level (log2)
4.5
5.5
6.5
7.5
20
30
40
50
60
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.9135)
MKL1 | 1559902_at
Expression level (log2)
4.5
5.5
6.5
7.5
20
30
40
50
60
Age
Smoker
non−smoker


set: Trachea (p=0.8523)
MKL1 | 1559902_at
Expression level (log2)
4.5
5.5
6.5
7.5
20
30
40
50
60
Age
Smoker
non−smoker


set: Macrophage1 (p=0.1826)
MKL1 | 1559902_at
Expression level (log2)
4.5
5.5
6.5
7.5
20
30
40
50
60
Age
Smoker
non−smoker


set: Macrophage2 (p=0.6848)
MKL1 | 1559902_at
Expression level (log2)
4.5
5.5
6.5
7.5
20
30
40
50
60
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: MKL1 | 1559902_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.3165
p=0.9419
p=0.4521
p=0.6561
p=0.4289
p=0.1143
p=0.5032
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2